As a general rule, you'll find that the colors interpolated in RGB tend toward muddy and gray, unless you break the color ramp into multiple stops. You can experiment with different color ramps, or stick to ramps that emphasize hue or saturation (by using HSL). Or you can rely on experts by using the built-in D3 functions for color ramps that are proven to be easier for a reader to distinguish, which we'll look at now.

DISCRETE COLORS

Oftentimes, we use color ramps to try to map colors to categorical elements. It's better to use the discrete color scales available in D3 for this purpose. The popularity of these scales is the reason why so many D3 examples have the same palette. To get started, we need to use a new D3 scale, `d3.scale.category10`, which is built to map categorical values to particular colors. It works like a quantizing scale where you can't change the domain, because the domain is already defined as 10 highly distinct colors. Instead, you instantiate your scale with the values you want mapped to those colors. In our case, we want to distinguish the various regions in our dataset, which consists of the top eight FIFA teams from the 2010 World Cup, representing four global regions. We want to represent these as different colors, and to do so, we need to create a scale with those values in an array.

```
function buttonClick(datapoint) {
    var maxValue = d3.max(incomingData, function(el) {
        return parseFloat(el[datapoint ]);
    });
    var tenColorScale = d3.scale.category10(
        ["UEFA", "CONMEBOL", "CAF",  "AFC"]);
    var radiusScale = d3.scale.linear().domain([0,maxValue]).range([2,20]);
    d3.selectAll("g.overallG").select("circle").transition().duration(1000)
        .style("fill", function(p) {return tenColorScale(p.region)})
        .attr("r", function(p) {return radiusScale(p[datapoint ])});
};
```

The application of this scale is visible when we click one of our buttons, which now resizes the circles as it always has, but also applies one of these distinct colors to each team (figure 3.16).

COLOR RAMPS FOR NUMERICAL DATA

Another option is to use color schemes based on the work of Cynthia Brewer, who has led the way in defining effective color use in cartography. Helpfully, d3js.org provides

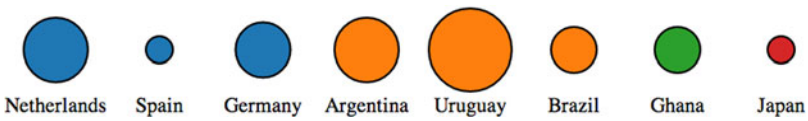


Figure 3.16 Application of the `category10` scale in D3 assigns distinct colors to each class applied, in this case, the four regions in your dataset.